

02-05-2025

ATT:

$$= E[Y(1) - Y(0) | D=1]$$

$$= E[Y(1) | D=1] - E[Y(0) | D=1]$$

$$\hookrightarrow y_i = D_i y_i(1) + (1 - D_i) y_i(0)$$

$$= y_i(0) + D_i (y_i(1) - y_i(0))$$

Experiment:

$$(y(1), y(0)) \perp\!\!\!\perp D$$

$\hookrightarrow$ fully independent of D (random assignment)

ATT:

$\hookrightarrow$ main challenge: $E[y(0) | D=1] \overset{\substack{\text{true w/} \\ \text{random assign}}}{=} E[y(0) | D=0]$

$$\hookrightarrow \text{ATT} = E[y | D=1] - E[y | D=0]$$

Relationship to Regression:

① model for untreated potential outcomes

$$\textcircled{2} y_i(0) = \beta_0 + e; E[e] = 0, E[e | D] = 0$$

$\hookrightarrow$ treatment effect homogeneity assumption:

$$TE_i = y_i(1) - y_i(0) = \alpha$$

$$y_i(0) + D_i(y_i(1) - y_i(0)) = y_i$$

$$y_i = \beta_0 + e_i + \alpha D_i$$

$$= \beta_0 + \alpha D_i + e_i \text{ s.t. } E[e_i | D] = 0$$

$$E[y_i | D] = \beta_0 + \alpha D \text{ (linear expectation)}$$

Fully Saturated Regression:

↳ all discrete variables w/ all possible interactions included

$$\hookrightarrow E[y | D=1] = \beta_0 + \alpha$$

$$- E[y | D=0] = \beta_0 = \alpha$$

↳ linear by construction

Unconfoundedness:

$$(y(1), y(0)) \perp\!\!\!\perp D \mid X$$

$$ATT = E[\underbrace{E[y(1) - y(0) | X, D=1]}_{ATT(X)} | D=1]$$

$$= E[E[y(1) | X, D=1] - E[y(0) | X, D=1] | D=1]$$

$$= E[E[y(1) | X, D=1] - E[y(0) | X, D=0] | D=1]$$

$$= E[E[y | X, D=1] - E[y | X, D=0] | D=1]$$

↳ ATT is nonparametrically identified

$$E[E[y|x, D=1] - E[y|x, D=0] | D=1]$$

↳ Partial effect of D given x (inside)

↳ Average partial effect (outside)

$$E[E[y|x, D=1] - E[y|x, D=0] | D=1]$$

$$= E[y | D=1] - E[E[y | D=0] | D=1]$$

Assumptions

- ① model for untreated potential outcomes
 $y_i(0) = x_i^T \beta + e$; $E[e|x] = 0$, $E[e|x, D] = 0$
 ↳ by unconfoundedness
- ② treatment effect homogeneity assumption:
 $TE_i = y_i(1) - y_i(0) = \alpha$

$$y_i = y_i(0) + D_i(y_i(1) - y_i(0))$$

$$= \alpha D_i + x_i^T \beta + e$$

$$E[y|x, D] = \alpha D + x^T \beta$$

Properties of Linear Projection :

$$\hookrightarrow \beta = \underset{b}{\operatorname{argmin}} E[(y - x^T b)^2]$$

$$= E[XX^T]^{-1} E[XY]$$

$\hookrightarrow$ Notation (Data Matrix)

$\rightarrow \underline{X} : n \times k$ matrix

$$\begin{pmatrix} x_{11} & x_{12} & \dots & x_{1k} \\ \vdots & \vdots & & \vdots \\ x_{n1} & \dots & \dots & x_{nk} \end{pmatrix} = \begin{pmatrix} x_1^T \\ \vdots \\ x_n^T \end{pmatrix}$$

$\rightarrow \underline{Y} : n \times 1$ vector

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$\rightarrow \underline{e} : n \times 1$ vector

$$\begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

Sampling:

$\hookrightarrow$ Simple random: $\{(y_i, x_i)\}_{i=1}^n$ that are i.i.d

$\rightarrow$ independent: one draw doesn't depend on another

$\rightarrow$ identically dist: all hold same distribution

$\hookrightarrow$ clustered: units within same cluster maybe not independent of one another

Estimation:

$\hookrightarrow E[Y] : \hat{E}[Y] = \frac{1}{n} \sum_{i=1}^n y_i$

$\hookrightarrow E[XX^T] : \hat{E}[XX^T] = \frac{1}{n} \sum_{i=1}^n x_i x_i^T$

$\hookrightarrow E[XY] : \hat{E}[XY] = \frac{1}{n} \sum_{i=1}^n x_i y_i$

$\hookrightarrow$ Plug-in Principle

$\rightarrow \beta = g(E[XX^T]^{-1} E[XY])$ s.t. $g(A, b) = A^{-1} b$

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i x_i^T^{-1} \frac{1}{n} \sum_{i=1}^n x_i y_i$$

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Working w/

Data Matrices

$$\hat{\beta} = (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y}$$